GraphQL Presentation

CSC 4160 Presentation

Topic: GraphQL

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Arjan Ellingson

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Agenda

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Intro - What is GraphQL?

GraphQL (you may see gql) may is an API method that allows for communication between a client or user and a server.

GraphQL is an alternative to the more common `REST` format used in web development.

Calls are executed from files with the extension `.gql` or `.graphql`.

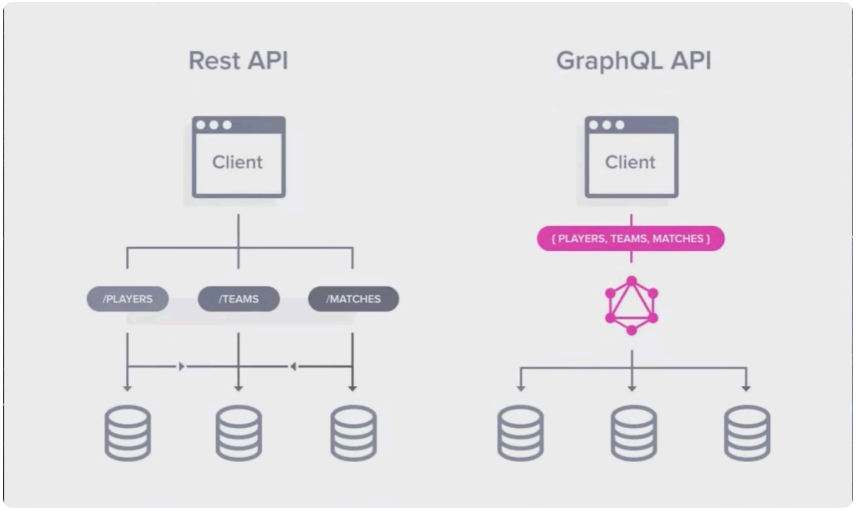
It is not limited to a specific database or backend language.

Brief History

- started by Facebook in 2012, aiming to becoming open source by 2015
- 2018: GraphQL was adopted by GraphQL Foundation, with parent Linux Foundation[1]
- Currently: APIs of companies like Facebook, GitHub (`gh cli`), Yelp, and Shopify utilize gql format

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The Alternative: REST



GraphQL combines each of REST's endpoints into one endpoint[2].

The Problem: Rest often **overfetches** or **underfetches** the data required.

The Alternative: REST (cont.)

REST

`GET /artists/42`

```
{
  "id": "42",
  "label": "Ada Lovelace",
```

GraphQL

`/graphql`

```
query {
  artist(id: "42") {
    label
```

```
  "genre": "Electronic",
  "era": "2010s",
  "image":
"https://example.com/ada.jpg",
  "kind": "PERSON",
  "visible": true
}
```

```
    genre
    songs {
      title
      spotifyUrl
    }
  }
}
```

GraphQL File & Call

```
schema {
  query: Query
  mutation: Mutation
}
type Query {}
type Mutation {}
```

`Query`: *Reading/Fetching*

`Mutation`: *Creation*, *Updating*, *Deletion*

Together: All operations of **CRUD**. Every mutation and query is synonymous to endpoints in rest. Now only one endpoint exists: `/graphql`.

Notes: **Resolver must match query structure!!**, Also: `Subscriptions` not covered

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Itty Bitty Demo

http://localhost:5173/ and Postman Software

```
%%THIS FETCHES ALL ARTISTS PERFORMING AT 8123 FEST%%
{
  edges(type: "performed_at", targetLabel: "8123 Fest") {
    source { label genre }
```

```
    # target { label }
  }
}

%%THIS ADDS K.Flay as an Artist%%
mutation {
  addArtist(label: "K.Flay") {
    id
    label
  }
}
%%Collab between K.Flay and Pierce The Veil %%
%%Mutation here assumes artist id is returned as k-flay%%
mutation {
  addEdge(source: "k-flay", target: "pierce-the-veil", type: "collaborated",
directed: false, year: 2023) {
    id
    source { id label genre }
    target { id label genre }
  }
}
```

Potential Capstone Applications

- This current sprint (`sprint 3`), working on "Dockerizing" our Calendar View for public user input & output.
- This may involve introducing a better API, in which we could use GraphQL
- GraphQL fits extremely well with Svelte (heavy part of stack)
- The tooling makes E2E type safety extremely easy.

Summary

- GraphQL is an API methodology that is an alternative to rest. Queries can be stored and called from a `.graphql` file

- Strengths: Returns a fixed shape of data based on what the client inputs.
- Demo: Showed real-life application in personal project to fetch and mutate data
- Potentially useful for MustangHours capstone project - depends on data structure (tbd).

Sources & References

- https://graphql.org/learn/
- https://youtu.be/eIQh02xuVw4?si=ohYxPe4DaoiooymK
- https://youtu.be/Zg4XIpnLWQg?si=LY9nEBPZprEv5LQO
- https://youtu.be/ZQL7tL2S0oQ?si=E7Xd3Phe-GaF483y

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1. https://en.wikipedia.org/wiki/GraphQL ↩
2. https://www.youtube.com/watch?v=Zg4XIpnLWQg ↩